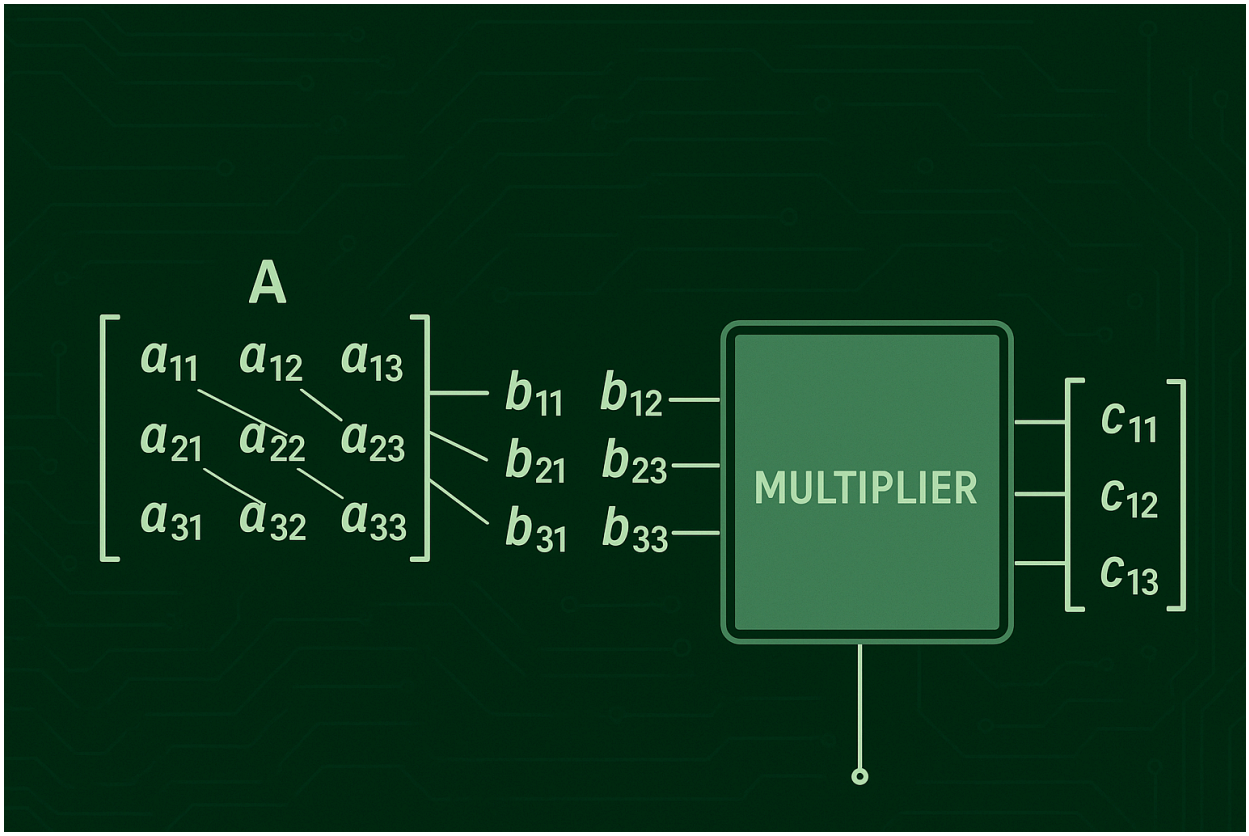


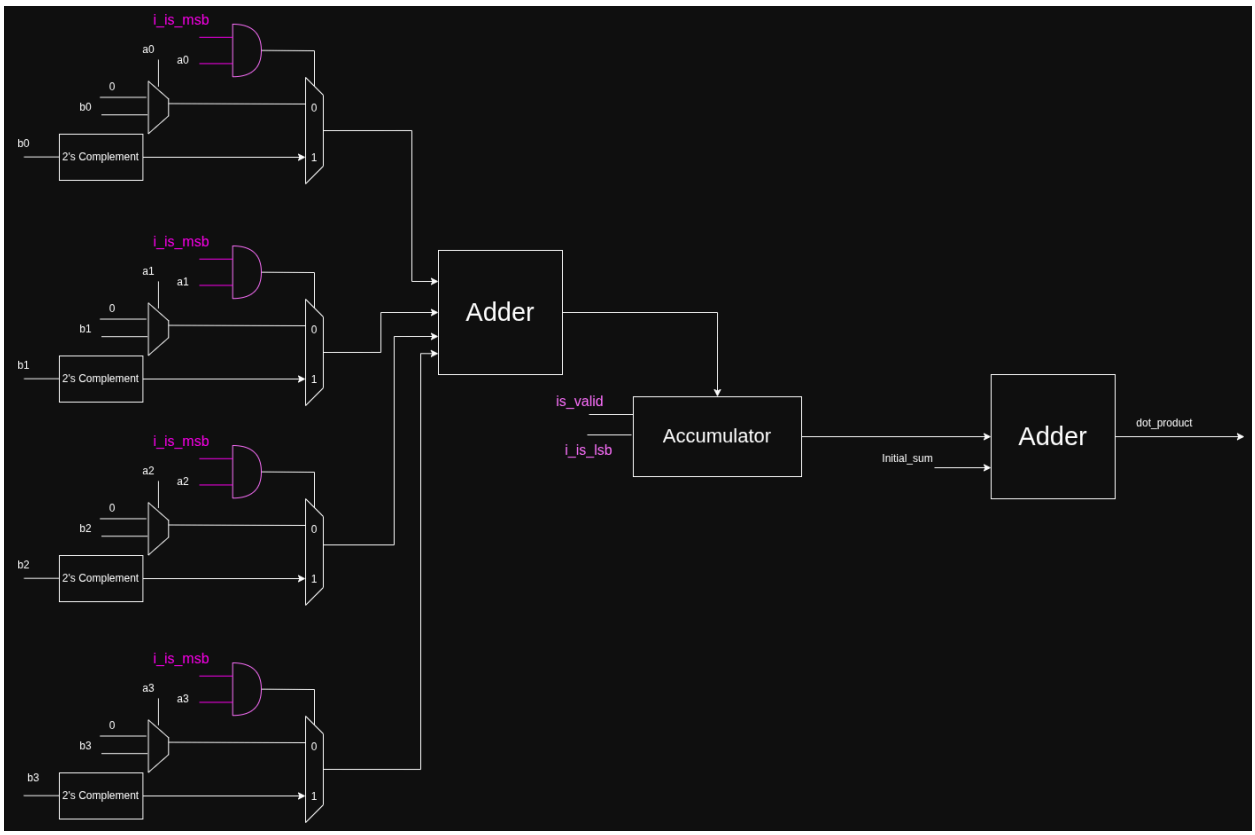
Matrix Multiplier

Erfan Falahati - 810102491

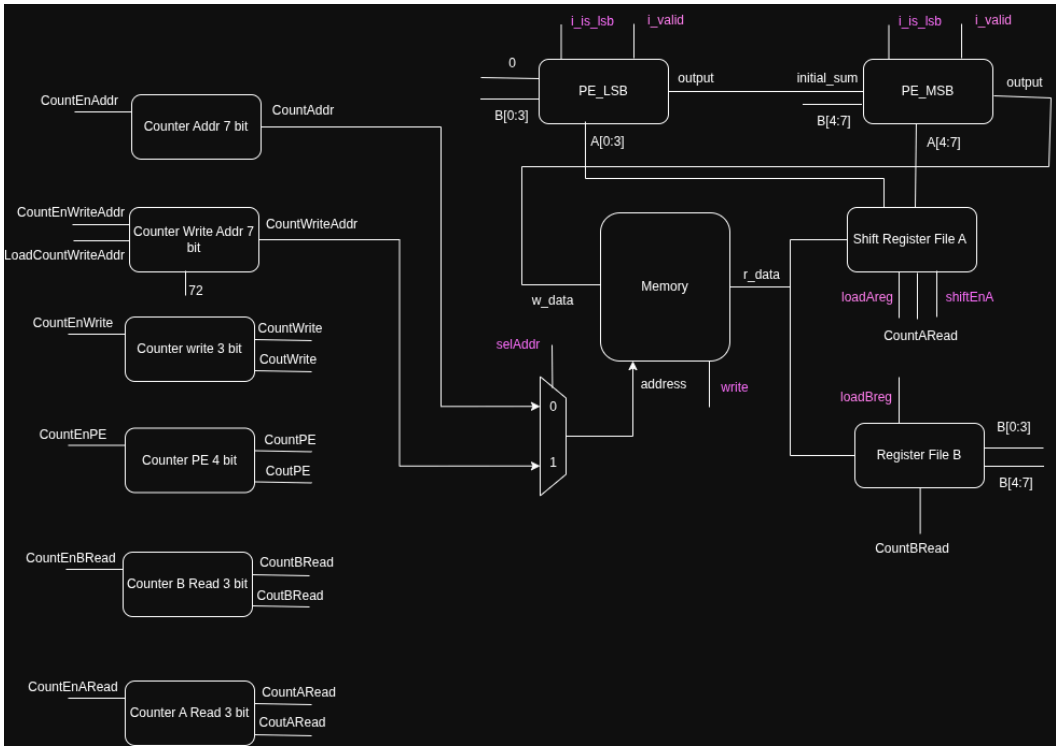
Sadegh Samadi - 810102588

CA#2

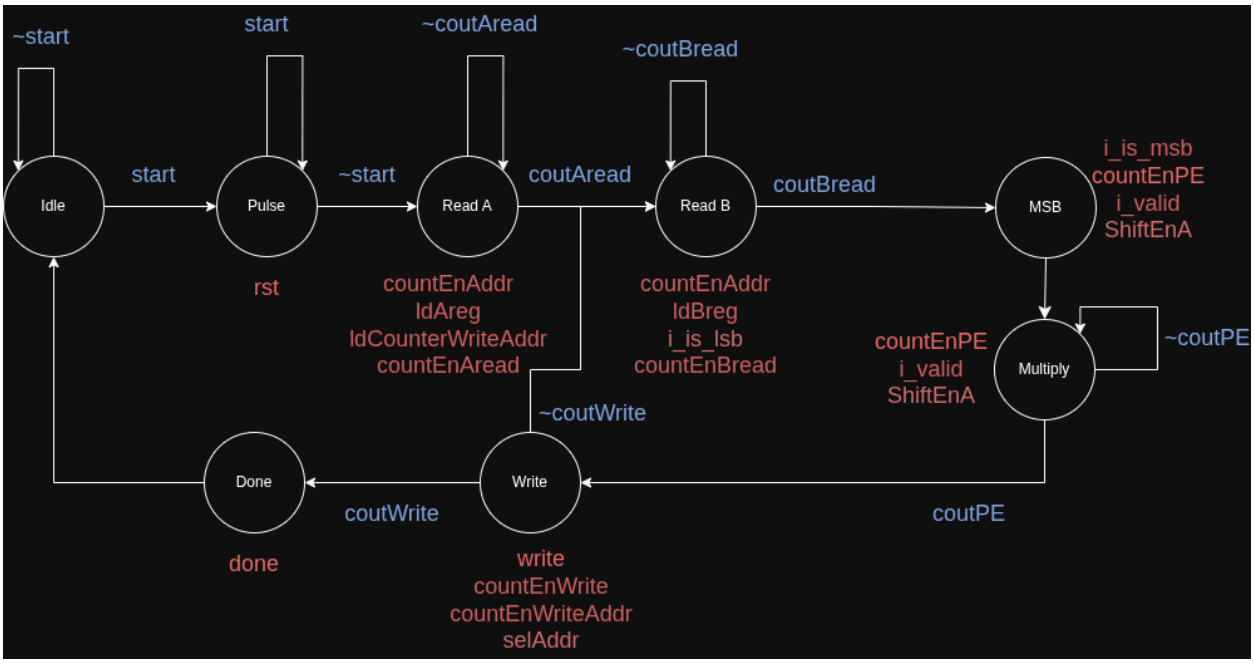
PE Data Path


i_is_msb
a0
a0
0
b0
b0
2's Complement
0
1
i_is_msb
a1
a1
0
b1
b1
2's Complement
0
1
i_is_msb
a2
a2
0
b2
b2
2's Complement
0
1
i_is_msb
a3
a3
0
b3
b3
2's Complement
0
1
Adder
is_valid
i_is_lsb
Accumulator
Initial_sum
Adder
dot_product

Matrix Vector Multiplier Data Path



Matrix Vector Multiplier Controller



Test Bench

```
module TestBenchMVM();
    reg clk;
    reg rst;
    reg start;
    wire done;

    initial clk = 1'b0;
    always #5 clk = ~clk;

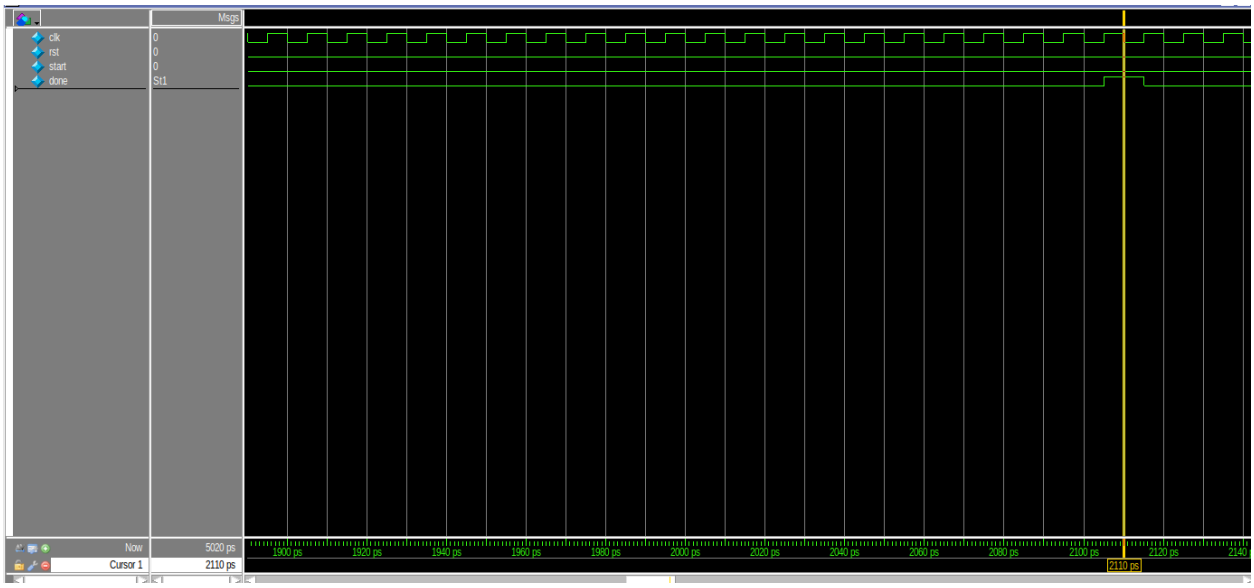
    Matrix_Multiplier #(
        .DATA_WIDTH(34),
        .ADDR_WIDTH(7),
        .DIMENSION(8),
        .MEMORY_NAME("input_memory (1).txt"),
        .MEMORY_SIZE(128),
        .WRITE_ADDR_START(72),
        .A_WIDTH(16),
        .B_WIDTH(16),
        .A_VEC_SIZE(8),
        .B_VEC_SIZE(8),
        .B_HALF_SIZE(4),
        .B_CONCAT_WIDTH(64)
    )UUT(
        .clk(clk),
        .rst(rst),
        .start(start),
        .done(done)
    );

    initial begin
        rst = 1'b1;
        #10;
        rst = 1'b0; start = 1'b1;
        #10;
        start = 1'b0;
        #5000;
        $stop;
    end

endmodule
```

Running TestCases

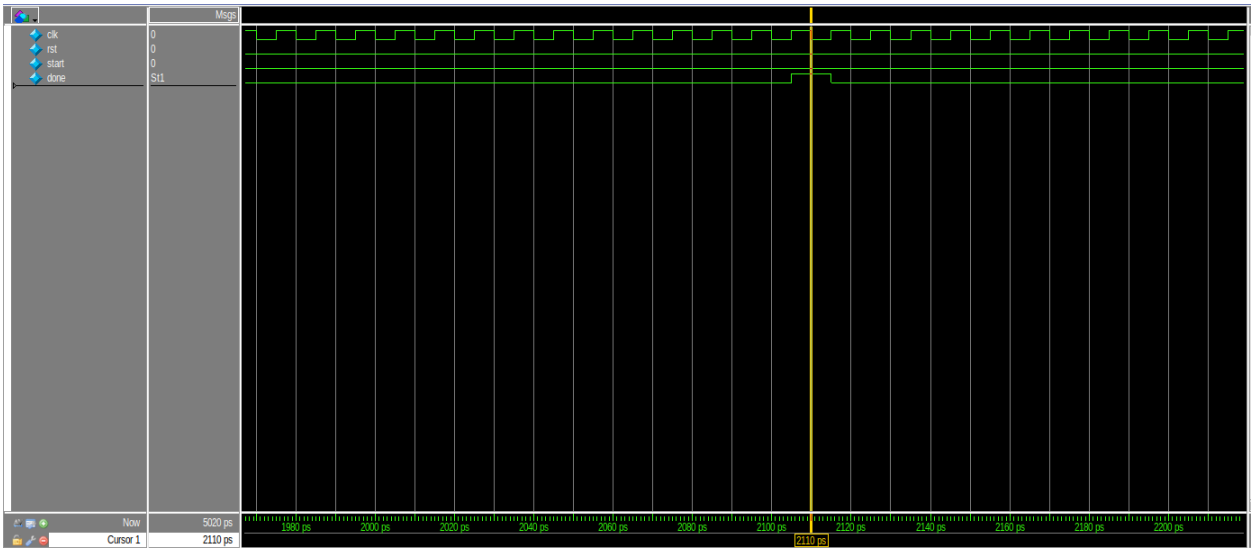
- input_memory (1)



Memory Values after done signal

```
 0 3ffffff9c 3ffffff9c 3ffffff9c 3ffffff9c 000000190 000000190 000000190 000000190 000000001 000000002 000000003 000000004 000000005 000000006 000000007 000000008 000000009 00000000a
18 00000000b 00000000c 00000000d 00000000e 00000000f 000000010 000000011 000000012 000000013 000000014 000000015 000000016 000000017 000000018 000000019 00000001a 00000001b 00000001c
36 00000001d 00000001e 00000001f 000000020 000000021 000000022 000000023 000000024 000000025 000000026 000000027 000000028 000000029 00000002a 00000002b 00000002c 00000002d 00000002e
54 00000002f 000000030 000000031 000000032 000000033 000000034 000000035 000000036 000000037 000000038 000000039 00000003a 00000003b 00000003c 00000003d 00000003e 00000003f 000000040
72 0000024b8 000004a38 000006fb8 000009538 00000bab8 00000e038 0000105b8 000012b38 xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx
```

- Input_mem (2)

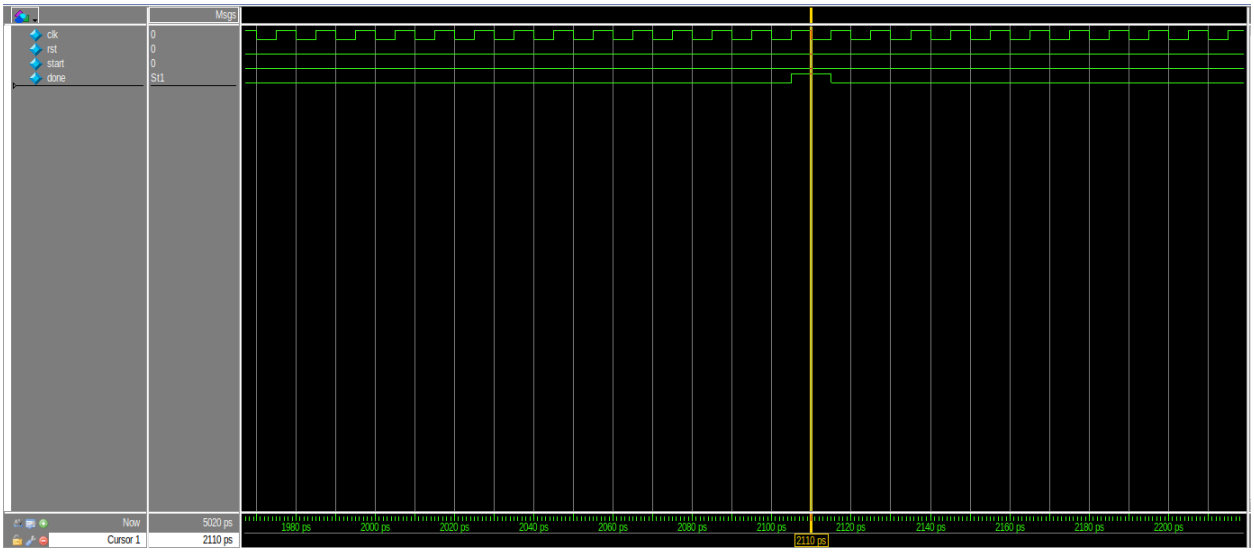


Memory Values after done signal

```
0  3ffffff97 3ffffff97 3ffffff97 3ffffff97 3fffffffe 3fffffffe 3fffffffe 3fffffffe 000000001 000000002 000000003 000000004 000000005 000000006 000000007 000000008 000000009 00000000a
18 00000000b 00000000c 00000000d 00000000e 00000000f 000000010 000000011 000000012 000000013 000000014 000000015 000000016 000000017 000000018 000000019 00000001a 00000001b 00000001c
36 00000001d 00000001e 00000001f 000000020 000000021 000000022 000000023 000000024 000000025 000000026 000000027 000000028 000000029 00000002a 00000002b 00000002c 00000002d 00000002e
54 00000002f 000000030 000000031 000000032 000000033 000000034 000000035 000000036 000000037 000000038 000000039 00000003a 00000003b 00000003c 00000003d 00000003e 00000003f 000000040
72 3fffffbb2 3fffee52 3fffe0f2 3ffffd392 3ffffc632 3ffffb8d2 3ffffab72 3fffff9e12 xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx
```

‘

- Input_mem (3)



Memory Values after done signal

```
0  3ffffff97 3fffffff1 3ffffffe9 00000001c 000000008 000000035 3fffffffe 000000029 000000000 000000002 3ffffffff 000000004 000000012 000000006 3ffffffff 000000008 000000071 00000000a
18 3ffffffff 00000000c 00000000d 00000000e 3ffffffff 000000010 3fffffff1 000000012 000000013 000000014 000000015 3fffffff1 000000017 000000018 3ffffffff 00000001a 3fffffffe 00000001c
36 00000001d 3ffffffff 3ffffffff 000000020 000000021 3fffffffe 000000023 000000004 000000005 000000006 000000002 3fffffff1 000000095 00000002a 00000000b 00000002c 00000002d 3ffffff97
54 3fffffff1 3ffffffe9 000000031 000000032 3ffffffff 000000034 000000035 3fffffff1 000000037 3fffffff1 000000039 00000003a 3fffffffe 00000003c 000000000 3fffffffe 00000003f 000000040
72 000000381 3ffffd858 0000006cb 0000007f6 3ffffeedb 3ffffac5a 3ffffea78 3fffff54f xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx xxxxxxxxx
```

‘